CV | CURRICULUM VITAE



PRITAM BOSE

+91-8910358939 | bosepritam762@gmail.com | https://pritam-kappa.vercel.app/
Surya Sen / Greater Kolkata, West Bengal
https://github.com/Pritam-Bose | https://www.linkedin.com/in/pritam-bose-63a3a33a2/

PROFESSIONAL PROFILE

Motivated B.Tech Computer Science (CSE) undergraduate at SVIST, Kolkata, with a strong foundation in software engineering and applied AI. Hands-on builder of full-stack and AI-powered projects spanning React front ends, FastAPI back ends and local large-language-model integration. Certified in Generative AI (AI-Variant, with training certification from EXCELR) and has completed an Ethical Hacking Basics internship (ICSS). Proficient across Python, JavaScript and React, with a keen interest in combining AI capabilities with robust software engineering practice.

CORE SKILLS

Languages: Python, JavaScript
Frameworks & Libraries: React.js, FastAPI
Web & Tools: HTML5, CSS3, Git, GitHub, JWT, bcrypt, Uvicorn, Edge-TTS, Web Audio API, Vercel, Railway, Kali Linux
Databases: PostgreSQL, MySQL, SQLite
Core Areas: Applied Generative AI, Full-Stack Web Development, API Design, Ethical Hacking Fundamentals
Security: Web Application Security, Vulnerability Assessment, Network Traffic Analysis, OWASP Top 10
Security Tools: Burp Suite, Nmap, Wireshark, Kali Linux

PROJECTS

Astha Provides | Business Website

Core Technologies: React.js, JavaScript, HTML5, CSS3

- Developed a responsive business website for Astha Provides to showcase IT services, security offerings, and support packages.
- Designed and implemented a modern UI with reusable components for services, gallery, testimonials, and contact sections.
- Integrated WhatsApp contact functionality to simplify client inquiry and lead generation.

Live Demo: ASTHA

Echo — Local AI-Powered Reflection System |

Technologies: Python, FastAPI, Llama.cpp (llama-cpp-python), Phi-3 Mini (GGUF), Edge-TTS, JavaScript (Web Audio API), HTML/CSS, Git

- Built a fully local, privacy-first AI reflection/journaling assistant running the Phi-3 Mini (GGUF) model via llama-cpp-python — no cloud inference dependency.
- Engineered a FastAPI backend to serve model inference and integrated Edge-TTS for natural voice output.
- Implemented audio capture/playback in the browser using the Web Audio API for a conversational, voice-driven experience.

Car Dashboard Web Application

Core Technologies: React.js, PostgreSQL, FastAPI

- Developed a web-based car dashboard interface using React for the frontend.
- Built backend APIs using FastAPI and integrated PostgreSQL for data management.
- Collaborative project developed jointly with a project partner.

Live Demo: CAR-SHOP-DASHBOARD

Cybersecurity Assessment Portfolio

Technologies: Kali Linux, Burpsuite, Wireshark, Nmap, Splunk

- Performed reconnaissance and web security testing using Burp Suite and Kali Linux.
- Investigated common web vulnerabilities and documented evidence, methodology and security impact.

Network Traffic Analysis & Monitoring

- Captured and analyzed TCP, HTTP, DNS and SSH traffic using Wireshark.
- Investigated packet flows, ports, TCP flags, request/response behavior and communication patterns.

Web Application Vulnerability Assessment & Remediation

- Identified and demonstrated SQL Injection, XSS and Broken Access Control in a controlled Flask lab.
- Implemented remediation and performed retesting to verify the vulnerabilities were addressed.

Api Security Assessment — crAPI

- Performed API security testing against a containerized crAPI application using Burp Suite and Docker.
- Analyzed authenticated API endpoints and demonstrated Broken Object Level Authorization (BOLA) by testing access to resources associated with different users..

SSH Authentication Security Assessment

- Performed SSH service reconnaissance using Nmap and analyzed authentication activity on a Metasploitable 2 laboratory system.
- Investigated failed and successful SSH authentication events using /var/log/auth.log and examined authentication activity based on usernames, source IP addresses and timestamps.

Security Monitoring & Log Analysis — Splunk

- Ingested SSH authentication logs into Splunk Enterprise and performed security-event analysis using Splunk Processing Language (SPL)**.**
- Created searches to identify failed and successful authentication events and analyzed their occurrence together as part of a basic security monitoring workflow.

TRAINING CERTIFICATIONS

- Generative AI — Training Certification, EXCELR
- Ethical Hacking Basics — Certification (3rd rank)

INTERNSHIP CERTIFICATIONS

- Generative AI — Internship Certification, AI-Variant
- Cyber Security & Ethical Hacking —Internship Certification, RABTECH ACADEMY

EDUCATION

B.Tech, Computer Science & Engineering | Swami Vivekananda Institute of Science and Technology (SVIST)

Expected 2027

Dakshin Gobindapur, Sonarpur, Kolkata, West Bengal, India | final Year undergraduate

ADDITIONAL INFORMATION

- Participated in a collegiate hackathon, applying rapid prototyping and problem-solving skills under time constraints.
- Active project builder with a public GitHub portfolio showcasing full-stack and AI-integrated applications.